Unit 2 FT 2022_Q5

Examiner only

5. Three images have been combined below to show colliding galaxy clusters 5 billion light years from Earth.

- The background is a Hubble Space Telescope image taken using visible light.
- The blue area is an X-ray image from the Chandra Space Observatory.
- The red area is an image from a radio telescope.

(a) (i) State **one** advantage of using space-based telescopes to produce images of objects in space. [1]

....

....

(ii) State **one** disadvantage of using space-based telescopes to produce images of objects in space. [1]

....

....

(b) Complete the following sentence by underlining the correct word in the brackets.

It takes radio waves 5 billion (**seconds / hours / days / years**) to travel from the colliding galaxy clusters to reach the Earth. [1]

Examiner only

(c) The image is taken using X-rays, radio waves and visible light.

(i) Which of these waves has the highest frequency? .... [1]

(ii) Which of these waves has the longest wavelength? .... [1]

(iii) Name **two** other types of electromagnetic waves that could be used to take images of galaxies. [2]

.... and ....

(d) The speed of electromagnetic waves through space is 300 000 000 m/s. Some radio telescopes on Earth detect radio waves which have wavelengths between 2 and 10 m.

Use the equation

$$\text{frequency} = \frac{\text{wave speed}}{\text{wavelength}}$$

to calculate the **maximum** frequency of radio waves detected. [2]

frequency = ... Hz

(e) Images of the Universe can be used to support the current theory of the origin of the Universe.

State the name of this theory. [1]

....

10